

A Conserved Histidine in the RNA Sensor RIG-I Controls Immune Tolerance to N1-2'O-Methylated Self RNA.

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Evidence abstract

The cytosolic helicase retinoic acid-inducible gene-I (RIG-I) initiates immune responses to most RNA viruses by detecting viral 5'-triphosphorylated RNA (pppRNA).

Although endogenous mRNA is also 5'-triphosphorylated, backbone modifications and the 5'-ppp-linked methylguanosine ((m7)G) cap prevent immunorecognition.

Here we show that the methylation status of endogenous capped mRNA at the 5'-terminal nucleotide (N1) was crucial to prevent RIG-I activation.

Moreover, we identified a single conserved amino acid (H830) in the RIG-I RNA binding pocket as the mediator of steric exclusion of N1-2'O-methylated RNA.

H830A alteration (RIG-I(H830A)) restored binding of N1-2'O-methylated pppRNA.

Consequently, endogenous mRNA activated the RIG-I(H830A) mutant but not wild-type RIG-I.

Similarly, knockdown of the endogenous N1-2'O-methyltransferase led to considerable RIG-I stimulation in the absence of exogenous stimuli.

Studies involving yellow-fever-virus-encoded 2'O-methyltransferase and RIG-I(H830A) revealed that viruses exploit this mechanism to escape RIG-I.

Our data reveal a new role for cap N1-2'O-methylation in RIG-I tolerance of self-RNA.

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Evidence checklist

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2. Although endogenous mRNA is also 5'-triphosphorylated, backbone modifications and the 5'-ppp-linked methylguanosine ((m7)G) cap prevent immunorecognition.
3. Here we show that the methylation status of endogenous capped mRNA at the 5'-terminal nucleotide (N1) was crucial to prevent RIG-I activation.
4. Moreover, we identified a single conserved amino acid (H830) in the RIG-I RNA binding pocket as the mediator of steric exclusion of N1-2'O-methylated RNA.
5. H830A alteration (RIG-I(H830A)) restored binding of N1-2'O-methylated pppRNA.
6. Consequently, endogenous mRNA activated the RIG-I(H830A) mutant but not wild-type RIG-I.
7. Similarly, knockdown of the endogenous N1-2'O-methyltransferase led to considerable RIG-I stimulation in the absence of exogenous stimuli.
8. Studies involving yellow-fever-virus-encoded 2'O-methyltransferase and RIG-I(H830A) revealed that viruses exploit this mechanism to escape RIG-I.
9. Our data reveal a new role for cap N1-2'O-methylation in RIG-I tolerance of self-RNA.

Source continuity

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